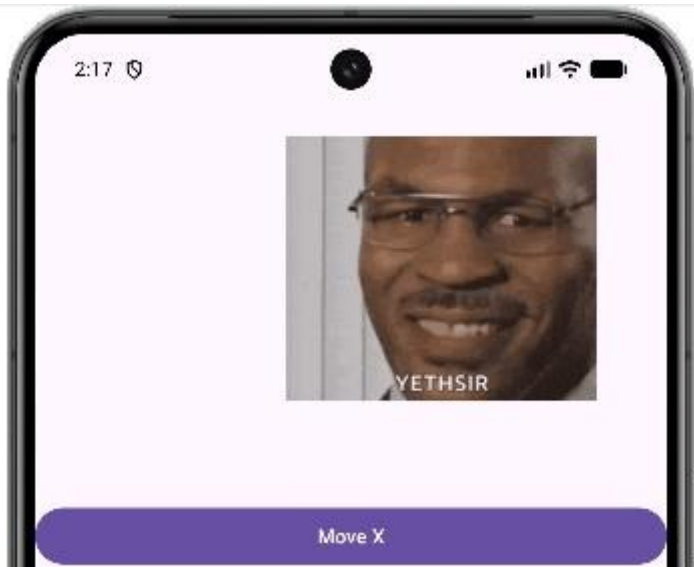


Create an android application to implement

1) Animation that moves the image from x direction to y direction.

X-axis -



Y-axis

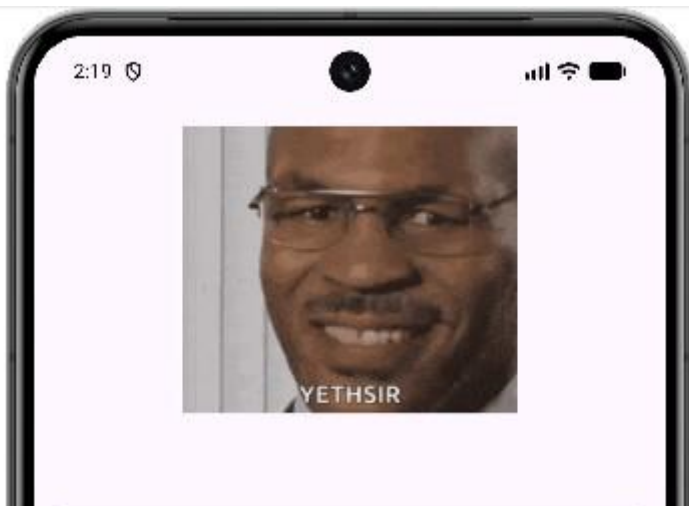


2) Scale Animation that expands and shrink image with respective button click.

Expand -



Shrink -



3) Rotate Animation that rotates Image (Clockwise and AntiClockwise).
Clockwise -



AntiClockwise -



4) Animation fades (In/Out) Image.

5) Animation slide (Up/Down) Image.

MainActivity.java

```
package com.example.animation;
```

```
import android.os.Bundle;
```

```
import android.view.animation.Animation;
```

```
import
```

```
android.view.animation.AnimationUtils;
```

```
import android.widget.Button;
```

```
import android.widget.ImageView;
```

```
import androidx.activity.EdgeToEdge;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import androidx.core.graphics.Insets;
```

```
import androidx.core.view.ViewCompat;
```

```
import androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    ImageView image;
```

```
    Button btnMoveX, btnMoveY, btnScaleUp, btnScaleDown, btnRotateCW, btnRotateACW,  
    btnFadeIn, btnFadeOut, btnSlideUp, btnSlideDown;
```

```

@Override
protected void onCreate(Bundle savedInstanceState)
{
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);
    image=findViewById(R.id.imageView);
    btnMoveX=findViewById(R.id.btnMoveX);
    btnMoveY=findViewById(R.id.btnMoveY);
    btnScaleUp=findViewById(R.id.btnScaleUp);
    btnScaleDown=findViewById(R.id.btnScaleDown);
    btnRotateCW=findViewById(R.id.btnRotateCW);
    btnRotateACW=findViewById(R.id.btnRotateACW);
    btnFadein=findViewById(R.id.btnFadein);
    btnFadeOut=findViewById(R.id.btnFadeOut);
    btnSlideDown=findViewById(R.id.btnSlideDown);
    btnSlideUp=findViewById(R.id.btnSlideUp);

    setButtonAnimation(btnMoveX, R.anim.move_x);
    setButtonAnimation(btnMoveY, R.anim.move_y);
    setButtonAnimation(btnScaleUp, R.anim.scale_up);
    setButtonAnimation(btnScaleDown, R.anim.scale_down);
    setButtonAnimation(btnRotateACW, R.anim.rotate_ac);
    setButtonAnimation(btnRotateCW, R.anim.rotate_cw);
    setButtonAnimation(btnFadein, R.anim.fade_in);
    setButtonAnimation(btnFadeOut, R.anim.fade_out);
    setButtonAnimation(btnSlideUp, R.anim.slide_up);
    setButtonAnimation(btnSlideDown, R.anim.slide_down);
}

private void setButtonAnimation(Button button, int animRes){
    button.setOnClickListener(v -> {
        Animation animation= AnimationUtils.loadAnimation(getApplicationContext(), animRes);
        image.startAnimation(animation);
    });
}
}

```

Activity_Main.xml -

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

```

```
<ImageView
    android:id="@+id/imageView"
    android:layout_width="match_parent"
    android:layout_height="300dp"
    android:src="@drawable/mtyson"
    android:scaleType="centerInside"
    android:layout_margin="16dp"
    android:contentDescription="Mike Tyson image for animation" />

<ScrollView
    android:layout_width="match_parent"
    android:layout_height="wrap_content">
    <LinearLayout
        android:layout_width="match_pare
        nt"
        android:layout_height="wrap_conte
        nt" android:orientation="vertical">

        <Button
            android:layout_width="match_pare
            nt"
            android:layout_height="wrap_conte
            nt" android:id="@+id/btnMoveX"
            android:text="Move X"/>

        <Button
            android:layout_width="match_pare
            nt"
            android:layout_height="wrap_conte
            nt" android:id="@+id/btnMoveY"
            android:text="Move Y"/>

        <Button
            android:layout_width="match_pare
            nt"
            android:layout_height="wrap_conte
            nt" android:id="@+id/btnScaleUp"
            android:text="Scale Up"/>

        <Button
            android:layout_width="match_pare
            nt"
            android:layout_height="wrap_conte
            nt" android:id="@+id/btnScaleDown"
            android:text="Scale Down"/>

        <Button
            android:layout_width="match_pare
            nt"
            android:layout_height="wrap_conte
            nt"
            android:id="@+id/btnRotateCW"
            android:text="Rotate Clockwise"/>

        <Button
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:id="@+id/btnRotateACW"
            android:text="Rotate Anti Clockwise"/>

        <Button
            android:layout_width="match_pare
            nt"
            android:layout_height="wrap_conte
            nt" android:id="@+id/btnFadeIn"
            android:text="Fade In"/>
```

```
<Button  
    android:layout_width="match_pare  
nt"  
    android:layout_height="wrap_conte  
nt" android:id="@+id/btnFadeOut"  
    android:text="Fade Out"/>
```

```
<Button
    android:layout_width="match_pare
nt"
    android:layout_height="wrap_conte
nt" android:id="@+id/btnSlideUp"
    android:text="Slide Upward"/>

<Button
    android:layout_width="match_pare
nt"
    android:layout_height="wrap_conte
nt"
    android:id="@+id/btnSlideDown"
    android:text="Slide Downward"/>

</LinearLayout>
</ScrollView>

</LinearLayout>
```

anim (folder) -
fade_in.xml

-

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
<alpha xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromAlpha="0.0"
    android:toAlpha="1.0"
    android:duration="1000"
    android:fillAfter="true"
/>

</set>
```

fade_out.xml -

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
    <alpha xmlns:android="http://schemas.android.com/apk/res/android"
        android:fromAlpha="1.0"
        android:toAlpha="0.0"
        android:duration="1000"
        android:fillAfter="true"
    />

</set>
```

move_x.xml -

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
    <translate xmlns:android="http://schemas.android.com/apk/res/android"
        android:fromXDelta="0"
        android:toXDelta="200"
        android:duration="1000"
        android:fillAfter="true"
    />

</set>
```

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```
</set>
```

move_y.xml -

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
  <translate xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromYDelta="0"
    android:toYDelta="200"
    android:duration="1000"
    android:fillAfter="true"
  />
</set>
```

rotate_ac.xml -

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
  <rotate xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromDegrees="360"
    android:toDegrees="0"
    android:pivotX="50%"
    android:pivotY="50%"
    android:duration="1000"
    android:fillAfter="true"
  />
</set>
```

rotate_cw.xml -

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
  <rotate xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromDegrees="0"
    android:toDegrees="360"
    android:pivotX="50%"
    android:pivotY="50%"
    android:duration="1000"
    android:fillAfter="true"
  />
</set>
```

scale_down.xml -

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
  <scale xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromXScale="2.0"
    android:toXScale="1.0"
    android:fromYScale="2.0"
    android:toYScale="1.0"
    android:pivotX="50%"
  />
</set>
```

```
    android:pivotY="50%"
    android:duration="1000"
    android:fillAfter="true"
  />
```

</set>

Scale_up.xml -

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
  <scale xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromXScale="1.0"
    android:toXScale="2.0"
    android:fromYScale="1.0"
    android:toYScale="2.0"
    android:pivotX="50%"
    android:pivotY="50%"
    android:duration="1000"
    android:fillAfter="true"
  />
</set>
```

Slide_down.xml -

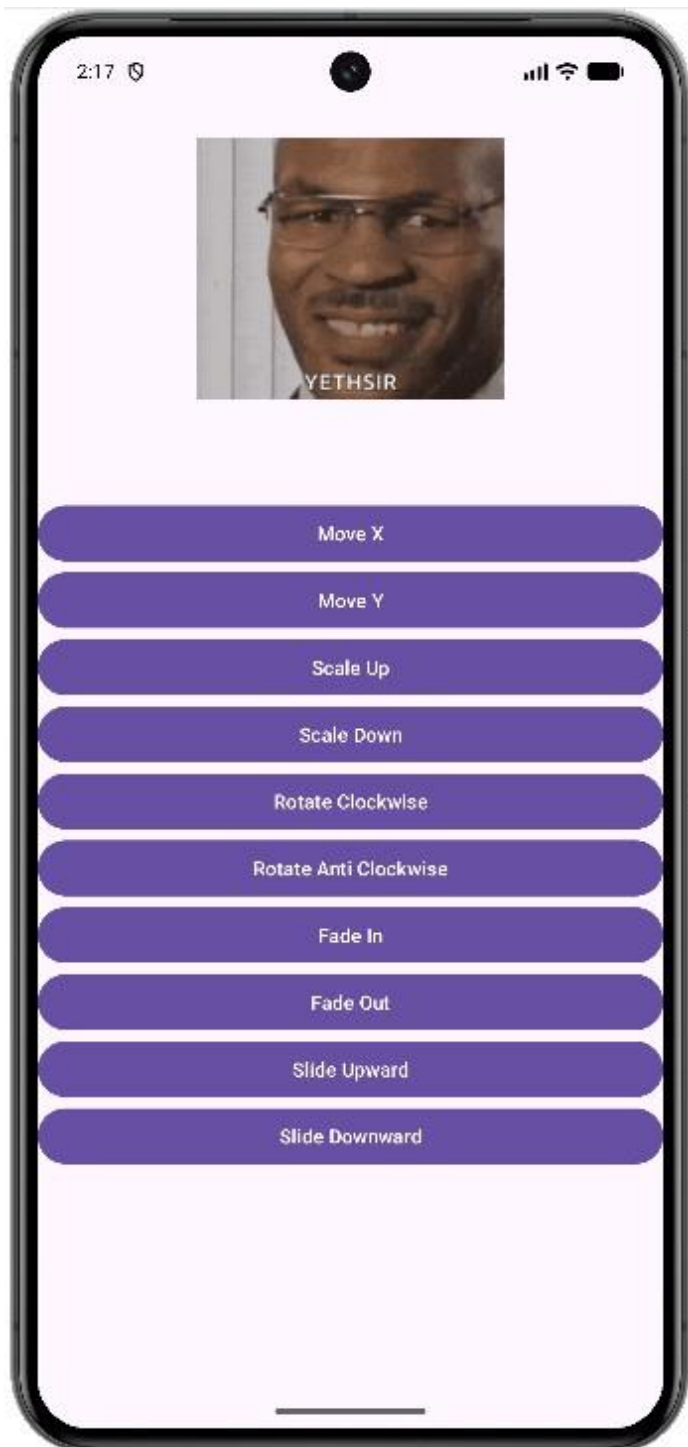
```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
  <translate xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromYDelta="0%"
    android:toYDelta="100%"
    android:duration="1000"
    android:fillAfter="true"
  />
</set>
```

Slide_up.xml -

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
  <translate xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromYDelta="100%"
    android:toYDelta="0%"
    android:duration="1000"
    android:fillAfter="true"
  />
</set>
```

</set>

Output -



Q2 Write an Android program to display an image, and play audio and video from the local resources.

Solution:

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp"
    tools:context=".MainActivity">

    <!-- Display Image -->
    <ImageView
        android:id="@+id/imageView"
        android:layout_width="250dp"
        android:layout_height="200dp"
        android:src="@drawable/image"
        android:contentDescription="Sample Image"
        android:scaleType="fitCenter" />

    <!-- Play Audio Button -->
    <Button
        android:id="@+id/btnPlayAudio"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Play Audio"
        android:layout_marginTop="20dp"/>

    <!-- VideoView -->
    <VideoView
        android:id="@+id/videoView"
        android:layout_width="match_parent"
        android:layout_height="200dp"
        android:layout_marginTop="20dp"
        />

    <!-- Play Video Button -->
    <Button
        android:id="@+id/btnPlayVideo"
        android:layout_width="wrap_content"
```

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```
android:layout_height="wrap_content"  
android:text="Play Video"
```

```
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.b10audiovideoplayer;
import
androidx.appcompat.app.AppCompatActivity;
import android.media.MediaPlayer;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import
android.widget.VideoView;
import android.widget.MediaController;

public class MainActivity extends AppCompatActivity

{ MediaPlayer mediaPlayer;
  VideoView videoView;
  Button btnPlayAudio, btnPlayVideo;

  @Override
  protected void onCreate(Bundle savedInstanceState)
  { super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    btnPlayAudio =
    findViewById(R.id.btnPlayAudio);
    btnPlayVideo =
    findViewById(R.id.btnPlayVideo); videoView =
    findViewById(R.id.videoView);

    // Audio player
    mediaPlayer = MediaPlayer.create(this, R.raw.song);
    btnPlayAudio.setOnClickListener(v -> {
        if (mediaPlayer.isPlaying())
        { mediaPlayer.pause();
          btnPlayAudio.setText("Play
          Audio");
        } else {
          mediaPlayer.start();
          btnPlayAudio.setText("Pause Audio");
        }
    });

    // Video player
    btnPlayVideo.setOnClickListener(v -> {
        String videoPath = "android.resource://" + getPackageName() + "/" + R.raw.vid;
```

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Uri uri = Uri.parse(videoPath);

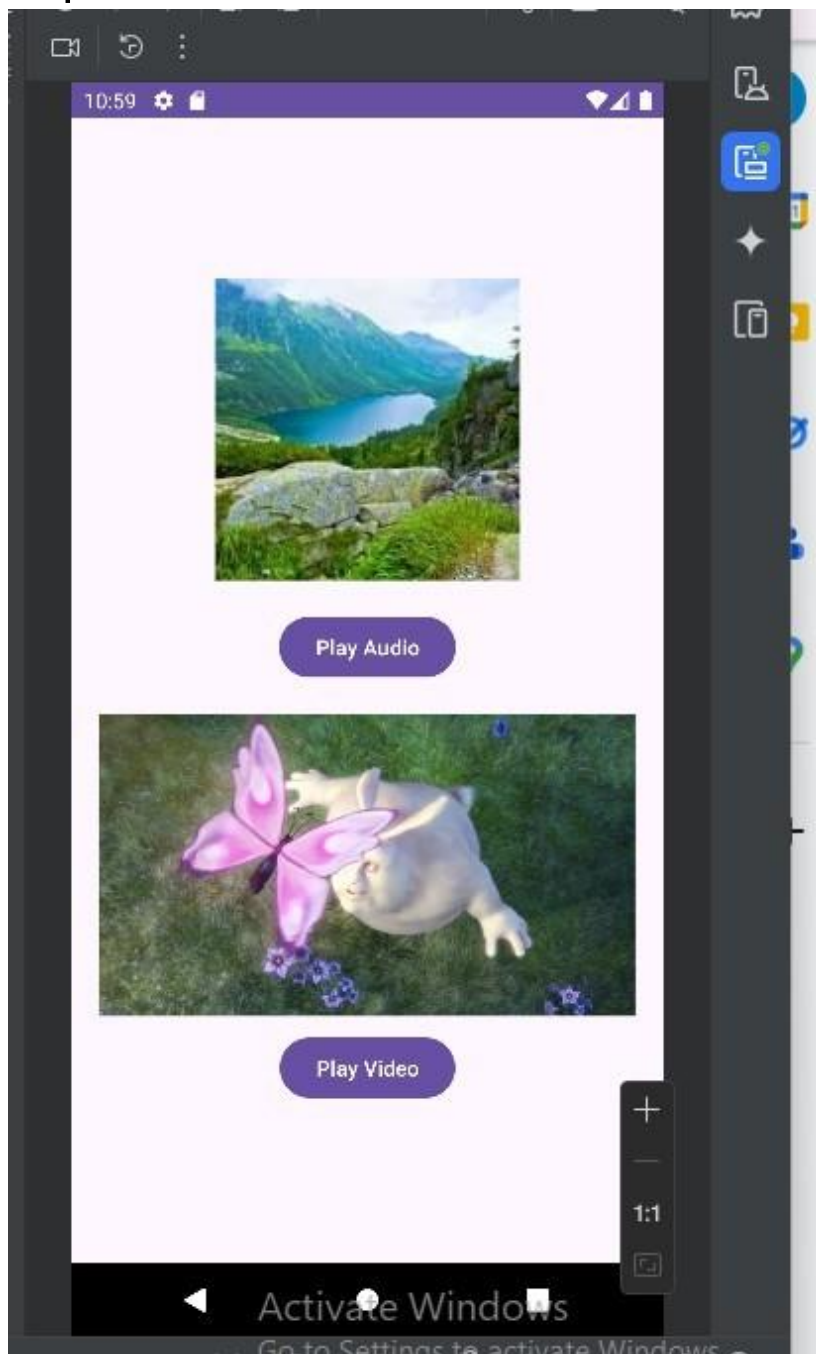
```
        videoView.setVideoURI(uri);

        // Media controls
        MediaController mediaController = new
        MediaController(this);
        videoView.setMediaController(mediaController);
        mediaController.setAnchorView(videoView);

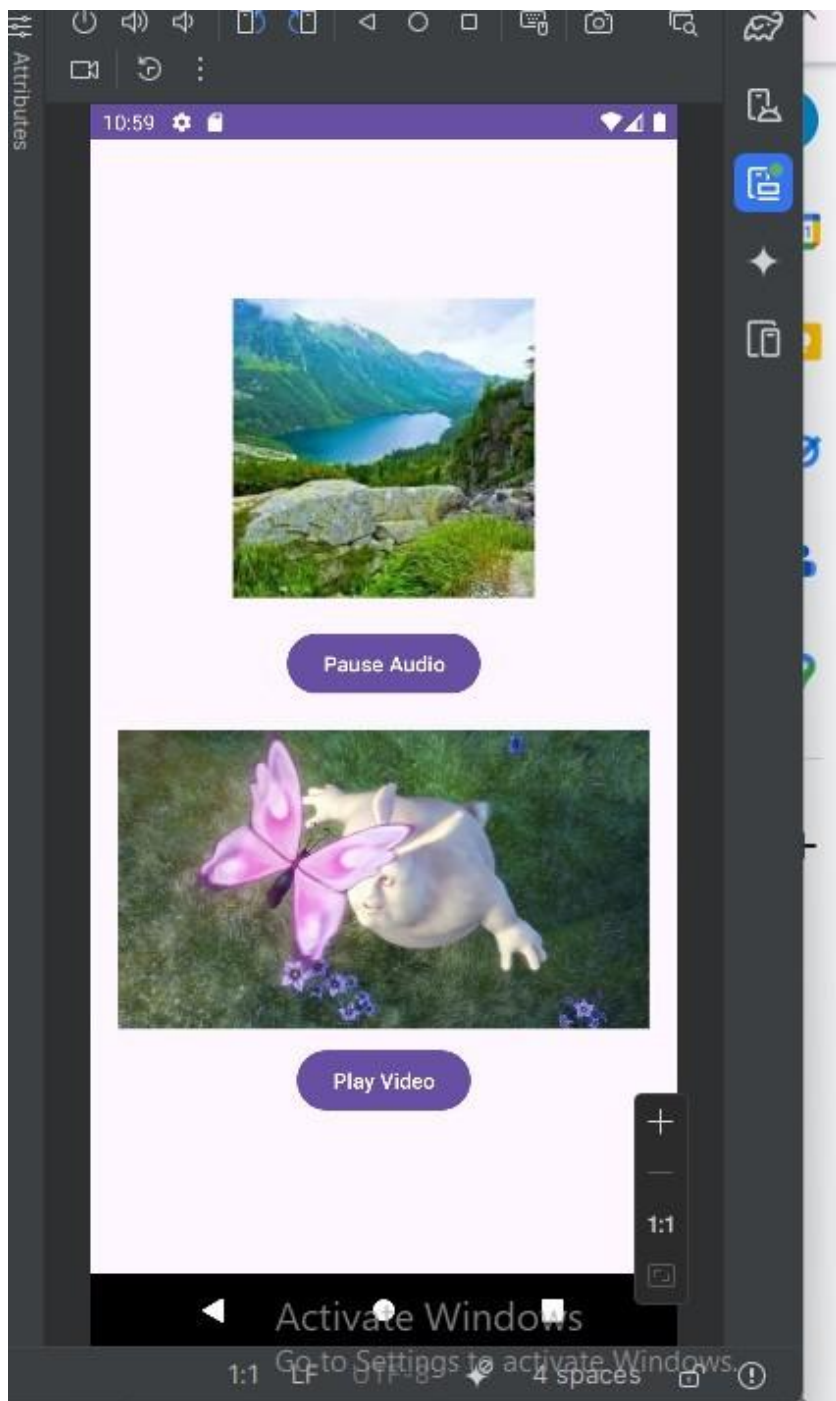
        videoView.start();
    });
}

@Override
protected void onDestroy()
{ super.onDestroy();
    if (mediaPlayer != null)
    {
        mediaPlayer.release();
        mediaPlayer = null;
    }
}
}
```

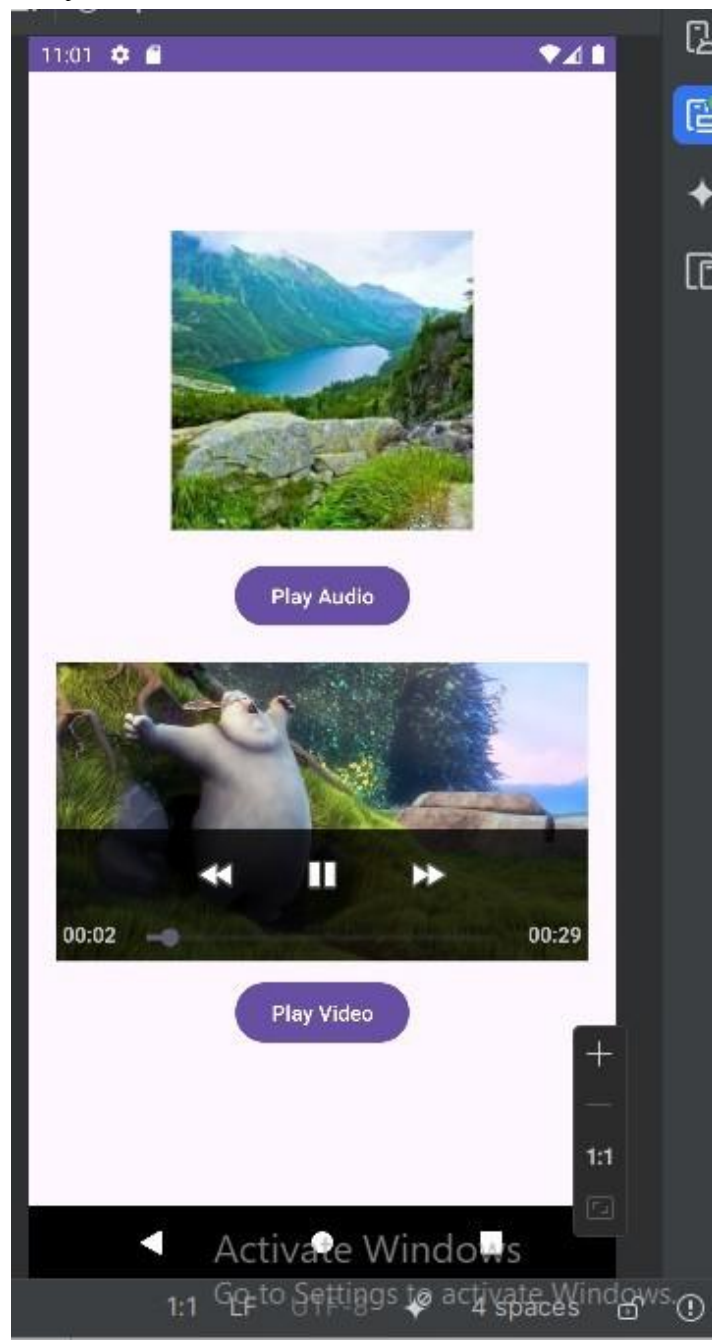
Output:



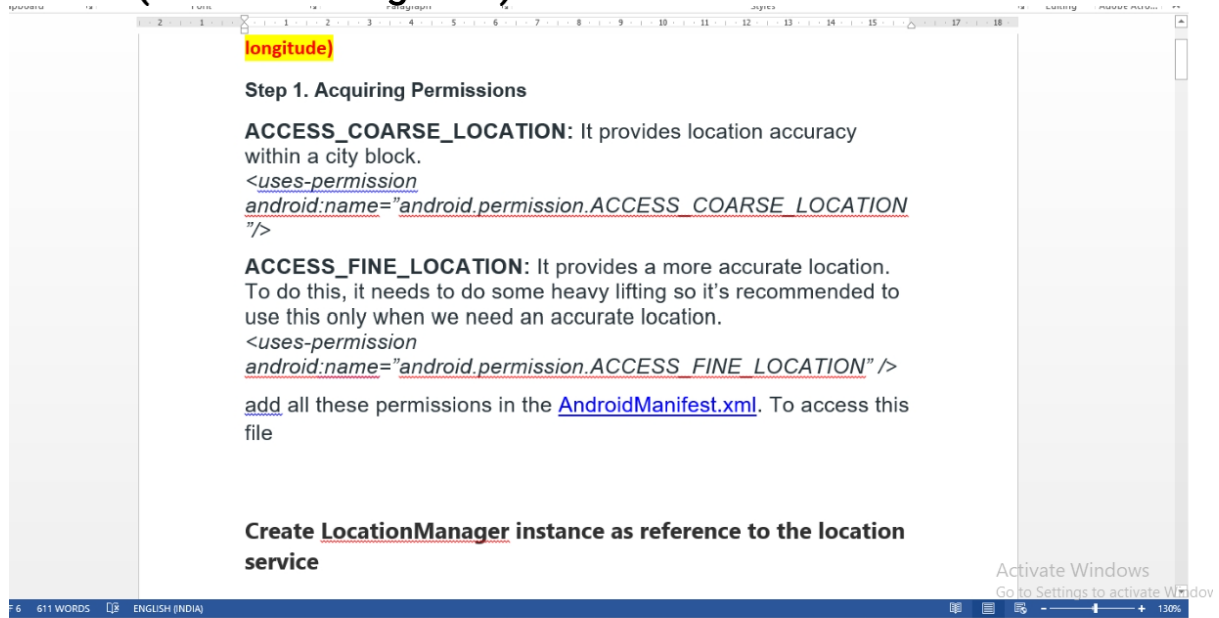
Play Audio:



Play audio:



Create an android application to display current location of the device (latitude & longitude).



longitude)

Step 1. Acquiring Permissions

ACCESS_COARSE_LOCATION: It provides location accuracy within a city block.
`<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION">`

ACCESS_FINE_LOCATION: It provides a more accurate location. To do this, it needs to do some heavy lifting so it's recommended to use this only when we need an accurate location.
`<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION">`

add all these permissions in the AndroidManifest.xml. To access this file

Create LocationManager instance as reference to the location service

Activate Windows
Go to Settings to activate Windows

6 611 WORDS ENGLISH (INDIA) 130%

Request current location from LocationManager

After creating the location service reference requested using requestLocationUpdates() method of LocationManager. For this function, we need to send the type of location provider, number of seconds, distance and the LocationListener object over which the location to be updated.

```
locationManager.requestLocationUpdates(LocationManager.GPS_PROVIDER, 0, 0, this);
```

LocationListener will be notified based on the distance interval specified or the number seconds.

- Android Get Address with Street Name, City for Location with Geocoding API

Geocoding

Geocoding is the process of converting the addresses (postal address) into geo coordinates as latitude and longitude. Reverse geocoding is converting a geo coordinate latitude and longitude to an address

Accessing the Geo Location for Lat and Long

Following class is the key element in accessing the latitude and longitude of the Android device. It implements the LocationListener and gets the location coordinate updates. We have designed our requirement not to be a continous update for

Android Get Address with Street Name, City for Location with Geocoding API

Geocoding

Geocoding is the process of converting the addresses (postal address) into geo coordinates as latitude and longitude. Reverse geocoding is converting a geo coordinate latitude and longitude to an address

Accessing the Geo Location for Lat and Long

Following class is the key element in accessing the latitude and longitude of the Android device. It implements the LocationListener and gets the location coordinate updates. We have designed our requirement not to be a continous update for location. On demand we will get the location coordinates.

```
locationManager = (LocationManager)
getSystemService(LOCATION_SERVICE);
// Permission Check:
if (ActivityCompat.checkSelfPermission(this,
Manifest.permission.ACCESS_FINE_LOCATION) !=
PackageManager.PERMISSION_GRANTED) {
    ActivityCompat.requestPermissions(this, new
String[]{Manifest.permission.ACCESS_FINE_LOCATION},
LOCATION_PERMISSION_REQUEST_CODE);
} else {
    requestLocationUpdates();
}
```

```
private final LocationListener locationListener = new LocationListener() {
    @Override
```

// This is called when the location changes. It extracts the latitude and longitude from the Location object and displays them in locationTextView. It then calls `fetchAddress()` to convert these coordinates into a readable address.

```
public void onLocationChanged(@NonNull Location location) {
    double latitude = location.getLatitude();
    double longitude = location.getLongitude();
    locationTextView.setText("Latitude: " + latitude + "\nLongitude: " +
longitude);
    fetchAddress(latitude, longitude);
}
```

```
@Override
public void onStatusChanged(String provider, int status, Bundle extras) {}
```

```
@Override
public void onProviderEnabled(@NonNull String provider) {}
```

```
@Override
public void onProviderDisabled(@NonNull String provider) {}
```

```

    }, 0, locationListener);
}

private final LocationListener locationListener = new LocationListener() {
    @Override

```

// This is called when the location changes. It extracts the latitude and longitude from the Location object and displays them in `locationTextView`. It then calls `fetchAddress()` to convert these coordinates into a readable address.

```

    public void onLocationChanged(@NonNull Location location) {
        double latitude = location.getLatitude();
        double longitude = location.getLongitude();
        locationTextView.setText("Latitude: " + latitude + "\nLongitude: " +
longitude);
        fetchAddress(latitude, longitude);
    }

```

```

    @Override
    public void onStatusChanged(String provider, int status, Bundle extras) {}

```

```

    @Override
    public void onProviderEnabled(@NonNull String provider) {
        double latitude = location.getLatitude();
        double longitude = location.getLongitude();
        locationTextView.setText("Latitude: " + latitude + "\nLongitude: " +
longitude);
        fetchAddress(latitude, longitude);
    }

```

```

    @Override
    public void onStatusChanged(String provider, int status, Bundle extras) {}

```

```

    @Override
    public void onProviderEnabled(@NonNull String provider) {}

```

```

    @Override
    public void onProviderDisabled(@NonNull String provider) {
        Toast.makeText(MainActivity.this, "Please enable GPS",
Toast.LENGTH_SHORT).show();
    }
}

```

```

private void fetchAddress(double latitude, double longitude) {
    Geocoder geocoder = new Geocoder(this, Locale.getDefault());
    try {
        List<Address> addresses = geocoder.getFromLocation(latitude,
longitude, 2);
        if (addresses != null && !addresses.isEmpty()) {
            String address = addresses.get(0).getAddressLine(0);
            locationTextView.append("\nAddress: " + address);
        } else {
            locationTextView.append("\nAddress not found");
        }
    } catch (IOException e) {
        e.printStackTrace();
        locationTextView.append("\nError fetching address");
    }
}

```

// • if the user grants the permission, it calls `requestLocationUpdates()` to start receiving location updates.

- If the user denies the permission, a Toast message ("Permission denied") is displayed.

// • if the user grants the permission, it calls `requestLocationUpdates()` to start receiving location updates.

- If the user denies the permission, a Toast message ("Permission denied") is displayed.

```

@Override
public void onRequestPermissionsResult(int requestCode, @NonNull
String[] permissions, @NonNull int[] grantResults) {
    super.onRequestPermissionsResult(requestCode, permissions,
grantResults);
    if (requestCode == LOCATION_PERMISSION_REQUEST_CODE) {
        if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
            requestLocationUpdates();
        } else {
            Toast.makeText(this, "Permission denied",
Toast.LENGTH_SHORT).show();
        }
    }
}

```

```
@Override
// It removes any location updates to stop location tracking and avoid
memory leaks.
protected void onDestroy() {
    super.onDestroy();
    if (locationManager != null) {
        locationManager.removeUpdates(locationListener);
    }
}
```

manifestFile

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.INTERNET"/>
    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"
/>
    <uses-permission
android:name="android.permission.ACCESS_COARSE_LOCATION" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Track location map"
```

API key.

To get one, follow the directions here:

<https://developers.google.com/maps/documentation/android-sdk/get-api-key>

Once you have your API key (it starts with "AIza"), define a new property in your project's `local.properties` file (e.g. `MAPS_API_KEY=AIza...`), and replace the

"YOUR_API_KEY" string in this file with "\${MAPS_API_KEY}".

-->

<meta-data

`android:name="com.google.android.geo.API_KEY"`

`android:value="AIzaSyDD7ozDaVm4TwWnuqgoAyTYFlUJjlZaec" />`

<activity

`android:name=".MapsActivity"`

`android:exported="true"`

`android:label="@string/title_activity_maps" >`

<intent-filter>

`<action android:name="android.intent.action.MAIN"/>`

`<category android:name="android.intent.category.LAUNCHER"/>`

</intent-filter>

</activity>

</application>

</manifest>

```
import java.util.Locale;

public class MapsActivity extends FragmentActivity implements
    OnMapReadyCallback {

    private static final int REQ_LOCATION = 101;
    private GoogleMap mMap;
    private FusedLocationProviderClient fused;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_maps);

        fused = LocationServices.getFusedLocationProviderClient(this);
        // Obtain the SupportMapFragment and get notified when the map is
        // ready to be used.
        SupportMapFragment mapFragment = (SupportMapFragment)
            getSupportFragmentManager()
                .findFragmentById(R.id.map);
        mapFragment.getMapAsync(this);
    }

    /**
     * Manipulates the map once available.
     * This callback is triggered when the map is ready to be used.
     * This is where we can add markers or lines, add listeners or move the
     * camera. In this case,
     * we just add a marker near Sydney, Australia.
     * If Google Play services is not installed on the device, the user
     * will be prompted to install
     * it inside the SupportMapFragment. This method will only be triggered
     * once the user has
     * installed Google Play services and returned to the app.
     */
    private void requestLocationOnce() {
        mMap = googleMap;
        requestLocationOnce();

        private void requestLocationOnce() {
            if (ActivityCompat.checkSelfPermission(this,
                Manifest.permission.ACCESS_FINE_LOCATION)
                != PackageManager.PERMISSION_GRANTED) {
                ActivityCompat.requestPermissions(this,
                    new String[]{Manifest.permission.ACCESS_FINE_LOCATION},
                    REQ_LOCATION);
                return;
            }
            fused.getLastLocation().addOnSuccessListener(this, location -> {
                if (location != null) {
                    double lat = location.getLatitude();
                    double lng = location.getLongitude();
                    LatLng myLocation = new LatLng(lat, lng);
                    Toast.makeText(MapsActivity.this,
                        "Latitude: " + lat + "\nLongitude: " + lng,
                        Toast.LENGTH_LONG).show();
                }
            });
        }
    }
}
```

```
// Get address
Geocoder geocoder = new Geocoder(MapsActivity.this,
Locale.getDefault());
try {
    List<Address> addresses = geocoder.getFromLocation(lat,
lng, 1);
    if (addresses != null && !addresses.isEmpty()) {
        String address =
addresses.get(0).getAddressLine(0);
        mMap.addMarker(new
MarkerOptions().position(myLocation).title(address));
        Toast.makeText(MapsActivity.this, address,
Toast.LENGTH_LONG).show();
    } else {
        mMap.addMarker(new
MarkerOptions().position(myLocation).title("No address found"));
    }
} catch (IOException e) {
    e.printStackTrace();
}

mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(myLocation, 15));
mMap.setMyLocationEnabled(true);
} else {
    Toast.makeText(MapsActivity.this, "Unable to get location",
Toast.LENGTH_SHORT).show();
}
});
}

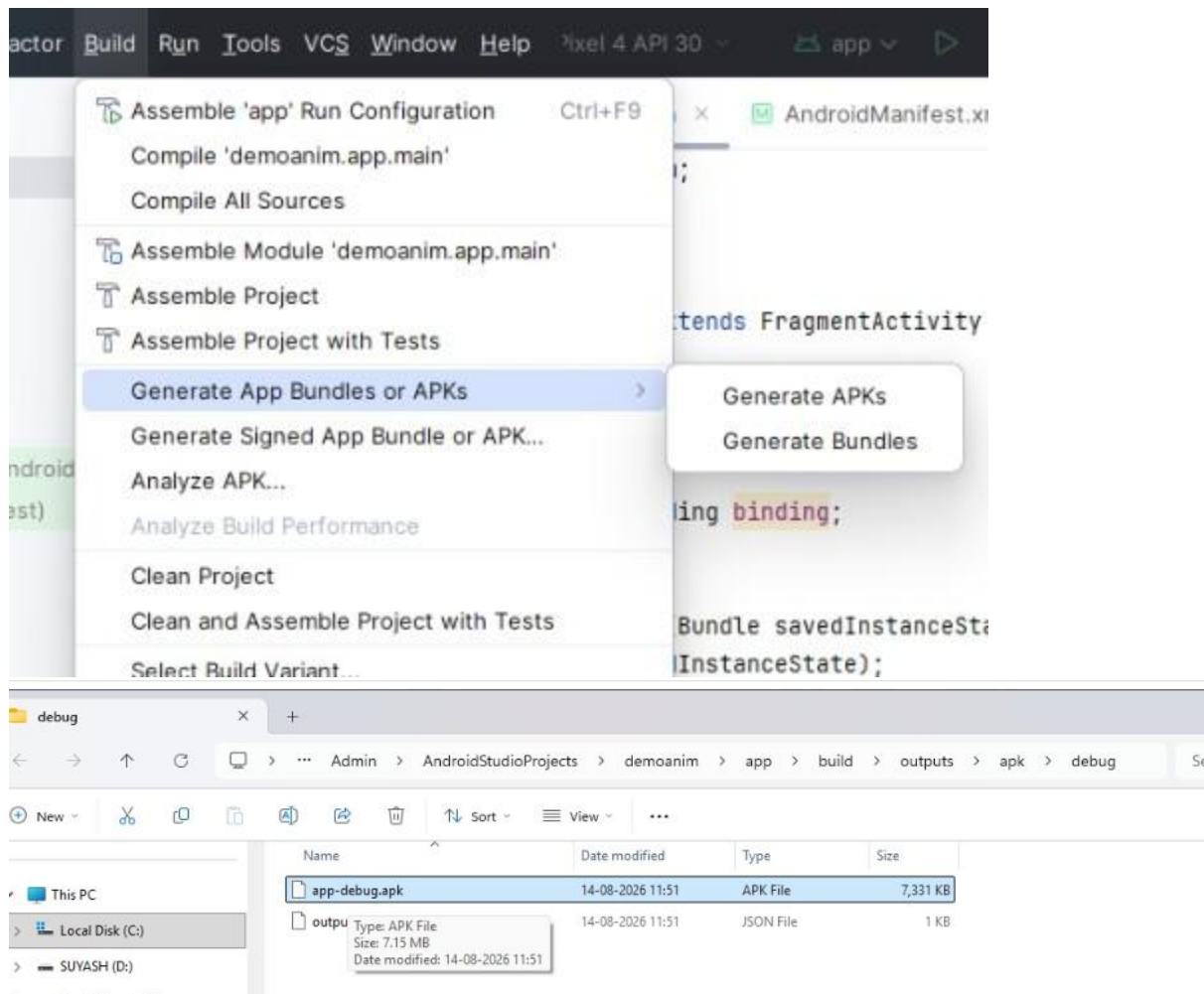
@Override public void onRequestPermissionsResult(int code, @NonNull
String[] perms, @NonNull int[] res) {
    super.onRequestPermissionsResult(code, perms, res);
    if (code == REQ_LOCATION && res.length > 0 && res[0] ==
```

gradle dependency

```
implementation ("com.google.android.gms:play-services-maps:19.2.0")
implementation ("com.google.android.gms:play-services-location:21.3.0")
```

<https://console.cloud.google.com/>

simple Android program that can track the device's location and display it on a map.



Create an android application to display current location of the device (latitude & longitude).

AndroidManifest.xml - `<?xml version="1.0" encoding="utf-8"?>`

`<manifest xmlns:android="http://schemas.android.com/apk/res/android"`
`xmlns:tools="http://schemas.android.com/tools">`

`<uses-permission android:name="android.permission.INTERNET" />`

`<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />`

`<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />`

```

<application
    android:allowBackup="true"
    android:dataExtractionRules="@xml/data_extraction_rules"
    android:fullBackupContent="@xml/backup_rules"
    android:icon="@mipmap/ic_launcher"
    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportRtl="true"
    android:theme="@style/Theme.Google_Activity">
    <activity
        android:name=".MainActivity"
        android:exported="true"
        android:windowSoftInputMode="adjustResize">
    </activity>
    <!--
        TODO: Before you run your application, you need a Google Maps API key.

```

To get one, follow the directions here:

<https://developers.google.com/maps/documentation/android-sdk/get-api-key>

Once you have your API key (it starts with "Alza"), define a new property in your project's local.properties file (e.g. MAPS_API_KEY=Aiza...), and replace the "YOUR_API_KEY" string in this file with "\${MAPS_API_KEY}".

```

-->
<meta-data
    android:name="com.google.android.geo.API_KEY"
    android:value="AlzaSyADFh7DaY8F8grTJuSLL2MQd-6EvH4w8" />

<activity
    android:name=".MapsActivity"
    android:exported="true"
    android:label="@string/title_activity_maps"
    android:windowSoftInputMode="adjustResize">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />

        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
</application>

</manifest>

```

MainActivity.java -

```

package com.example.google_activity;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);
    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
        return insets;
    });
}
```

MapsActivity.java -

```
package com.example.google_activity;
```

```
import androidx.annotation.NonNull;
import androidx.core.app.ActivityCompat;
import androidx.fragment.app.FragmentActivity;
```

```
import android.Manifest;
import android.content.pm.PackageManager;
import android.location.Address;
import android.location.Geocoder;
import android.os.Bundle;
import android.widget.Toast;
```

```
import com.google.android.gms.location.FusedLocationProviderClient;
import com.google.android.gms.location.LocationServices;
import com.google.android.gms.location.Priority;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;
import com.example.google_activity.databinding.ActivityMapsBinding;
import com.google.android.gms.tasks.OnSuccessListener;
```

```
import java.io.IOException;
import java.util.List;
import java.util.Locale;
```

```
public class MapsActivity extends FragmentActivity implements OnMapReadyCallback {
```

```
    private static final int REQ_LOCATION = 101;
    private GoogleMap mMap;
    private ActivityMapsBinding binding;
    private FusedLocationProviderClient fused;
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

    binding = ActivityMapsBinding.inflate(getLayoutInflater());
    setContentView(binding.getRoot());

    fused = LocationServices.getFusedLocationProviderClient(this);

    // Obtain the SupportMapFragment and get notified when the map is ready to be used.
    SupportMapFragment mapFragment = (SupportMapFragment) getSupportFragmentManager()
        .findFragmentById(R.id.map);
    if (mapFragment != null) {
        mapFragment.getMapAsync(this);
    }
}

@Override
public void onMapReady(GoogleMap googleMap) {
    mMap = googleMap;
    requestLocationOnce();
}

private void requestLocationOnce() {
    if (ActivityCompat.checkSelfPermission(this, Manifest.permission.ACCESS_FINE_LOCATION) !=
        PackageManager.PERMISSION_GRANTED &&
        ActivityCompat.checkSelfPermission(this,
            Manifest.permission.ACCESS_COARSE_LOCATION) != PackageManager.PERMISSION_GRANTED) {
        ActivityCompat.requestPermissions(this, new
            String[]{Manifest.permission.ACCESS_FINE_LOCATION, REQ_LOCATION});
        return;
    }

    fused.getCurrentLocation(Priority.PRIORITY_HIGH_ACCURACY,
        null).addOnSuccessListener(this, location -> {
        if (location != null) {
            double lat = location.getLatitude();
            double lng = location.getLongitude();
            LatLng myLocation = new LatLng(lat, lng);

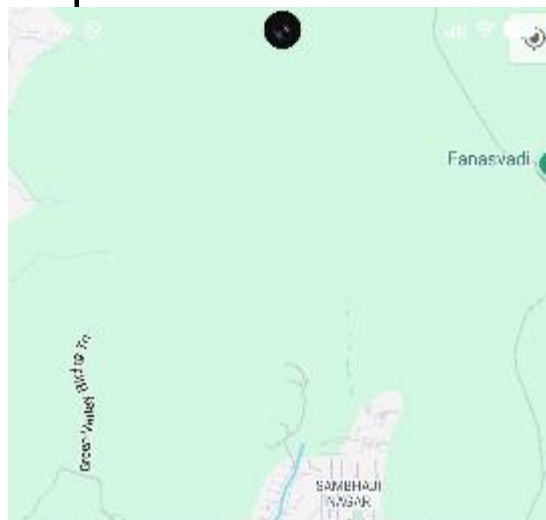
            // Get address
            Geocoder geocoder = new Geocoder(MapsActivity.this, Locale.getDefault());
            try {
                List<Address> addresses = geocoder.getFromLocation(lat, lng, 1);
                if (addresses != null && !addresses.isEmpty()) {
                    String address = addresses.get(0).getAddressLine(0);
                    mMap.addMarker(new MarkerOptions()
                        .position(myLocation)
                        .title(address)
                        .snippet("Lat: " + lat + ", Lng: " + lng));
                    Toast.makeText(MapsActivity.this, address, Toast.LENGTH_LONG).show();
                } else {
                    mMap.addMarker(new MarkerOptions()
                        .position(myLocation)
                        .title("No address found")
                        .snippet("Lat: " + lat + ", Lng: " + lng));
                }
            } catch (IOException e) {
```

```
        mMap.addMarker(new MarkerOptions()
            .position(myLocation)
            .title("Error fetching address")
            .snippet("Lat: " + lat + ", Lng: " + lng));
    }

    Toast.makeText(MapsActivity.this, "Latitude: " + lat + "\nLongitude: " + lng,
        Toast.LENGTH_LONG).show();
    mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(myLocation, 15));
    if (ActivityCompat.checkSelfPermission(MapsActivity.this,
        Manifest.permission.ACCESS_FINE_LOCATION) == PackageManager.PERMISSION_GRANTED) {
        mMap.setMyLocationEnabled(true);
    }
    else {
        Toast.makeText(MapsActivity.this, "Unable to get location. Ensure GPS is enabled.",
            Toast.LENGTH_SHORT).show();
    }
    });
}

@Override
public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions,
    @NonNull int[] grantResults) {
    super.onRequestPermissionsResult(requestCode, permissions, grantResults);
    if (requestCode == REQ_LOCATION) {
        if (grantResults.length > 0 && grantResults[0] == PackageManager.PERMISSION_GRANTED) {
            requestLocationOnce();
        }
        else {
            Toast.makeText(this, "Permission denied", Toast.LENGTH_SHORT).show();
        }
    }
}
}
```

Output -



, YMCA Marg, Sector 3A, CBD Belapur, Navi Mumbai, Maharashtra
at: 19.0294017, Lng: 73.0420283

